

DECLARATION

The author hereby declares that this thesis paper has been submitted as a requirement for the partial fulfillment of the B. Sc degree in Computer Science and Engineering. The work has not previously been accepted in substance for any other degree nor submitted in any previous application for any other degree. This dissertation is the result of my own work, except where otherwise stated. Other sources are acknowledged by explicit references to the authors.

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All praise and gratitude are due to the Almighty Creator, the source of all knowledge and wisdom, for bestowing upon us a portion of His boundless knowledge and granting us the strength, patience, and perseverance to successfully complete and submit our final year thesis within the stipulated time.

It is often said that there are two ways of spreading light: to be the candle itself or to be the mirror that reflects it. In the pursuit of knowledge, this work represents only a humble mirror, reflecting the illumination provided by the many guiding lights around us. This thesis would not have been possible without the continuous support, guidance, and encouragement of numerous individuals to whom we owe our sincere gratitude.

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DEDICATION

This thesis is completely dedicated to our cherished families, whose support, love, and never-ending sacrifices have been the strongest force during this trip. The faith they had in us made us chase our dreams with zeal and toughness.

Our teachers and mentors, who are considered the most respected, are the ones to whom we dedicate this work. The influence of their guidance, wisdom, and patience can be seen not only in this research but also in our development academically. They, through their commitment to the cultivation of knowledge and curiosity, have been the driving force behind our exploration of enterprise infrastructure security, behavioral intelligence, and robust digital boundaries.

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At the end, we dedicate this work to all organizations affected by data breaches and credential compromises. Let the research open up some paths to a world where software architecture is not only empowering, but also protecting and safeguarding rather than leaving enterprise boundaries exposed.

LETTER OF APPROVAL

This is to certify that the thesis titled “**AI-Driven Behavioral Anomaly Detection for Zero-Trust Role-Based Access Control in Monolithic Architecture**” submitted by Shafayeth Ahmed Chowdhury Academic ID: 0992220005101006, Nawrin Jahan Nancy Academic ID: 0992220005101005 has been examined and accepted as fulfilling the requirements for the degree of Computer Science & Engineering.

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ABSTRACT

Monolithic architectures remain widely utilized across business infrastructures due to their operational simplicity and rapid deployment capabilities. However, these tightly coupled systems face significant internal and external cybersecurity threats, as conventional perimeter security models fail to address credentials compromise or internal malicious behavior. Traditional Role-Based Access Control (RBAC) relies on static access credentials, which proves inadequate once an authenticated account exhibits malicious intent. To mitigate this security vulnerability, this thesis introduces an automated, intelligent security system that integrates machine learning directly into access controls using a dynamic Zero Trust Architecture (ZTA) framework.

The proposed framework introduces an automated real-time middleware interception layer that continuously screens incoming API user sessions using an unsupervised Isolation Forest machine learning model. This behavioral analysis vector evaluates user habits through eight critical behavioral dimensions, including login hour, query frequencies per minute, IP entropy, session duration, privilege mismatches, and data volume consumption rates. By capturing raw, multi-dimensional session data directly from a monolithic core system (built on Web, PHP/Laravel, and Python components), the architecture constructs real-time risk scores and dynamically maps them to real-time adaptive trust thresholds.

Experimental validation utilizing real-world and synthetic baseline datasets demonstrates the system's high reliability in isolating anomalous actions from legitimate sessions, successfully flags privilege escalation attempts, and counters data exfiltration indicators. The proposed architecture establishes that integrating AI-driven continuous verification into monolithic deployments provides a scalable, defensive solution that enforces proactive digital protection without requiring a complete shift to decentralized microservices.

Keywords: Zero Trust Architecture (ZTA), Role-Based Access Control (RBAC), Behavioral Anomaly Detection, Isolation Forest, Machine Learning in Cybersecurity, Monolithic Web Application Security, Continuous Authentication, Dynamic Access Control, User and Entity Behavior Analytics (UEBA), Real-Time Risk Scoring.

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Chapter 1

Introduction

1.1. Overview

This thesis proposes an AI-driven security framework that extends Role-Based Access Control (RBAC) into a Zero-Trust model for monolithic system architectures. RBAC, as formalized by Sandhu et al. (1996), assigns permissions according to predefined organizational roles but offers no mechanism for re-evaluating a user once a session has been authorized. Although monolithic systems are valued for their architectural simplicity, their tightly coupled internal design means that an attacker who compromises a single account can often move laterally through the application with comparatively little resistance. To close this gap, the study integrates behavioral anomaly detection so that authentication shifts from a single, binary event at login toward continuous, context-aware verification throughout the session.

This thesis proposes an AI-driven security framework that extends Role-Based Access Control (RBAC) into a Zero-Trust model for monolithic system architectures. RBAC, as formalized by Sandhu et al. (1996), assigns permissions according to predefined organizational roles but offers no mechanism for re-evaluating a user once a session has been authorized. Although monolithic systems are valued for their architectural simplicity, their tightly coupled internal design means that an attacker who compromises a single account can often move laterally through the application with comparatively little resistance. To close this gap, the study integrates behavioral anomaly detection so that authentication shifts from a single, binary event at login toward continuous, context-aware verification throughout the session.

The core of the research is the development of machine learning models, particularly the Isolation Forest algorithm introduced by Liu et al. (2008), to build a statistical baseline of normal user behavior. By analyzing temporal access patterns, request frequency, and transactional sequences, the proposed system is designed to flag subtle deviations that may indicate credential theft or insider misuse. Consistent with the Zero-Trust principle that no actor should be trusted by default regardless of network location or prior authentication (Rose et al., 2020), any detected anomaly is intended to trigger an automated response, such as step-up authentication or session revocation, independent of the user's assigned role.

Taken together, this research argues that AI-driven behavioral analytics can give monolithic systems a more resilient security posture without requiring a full architectural rewrite. In the proposed model, permissions are not granted purely on the basis of static role assignment; they are continuously re-evaluated against real-time behavioral evidence, narrowing the window in which a threat that has already bypassed perimeter defenses can operate undetected.

1.2. Problem Statement

Based on the gap analysis above, the problem addressed by this study can be summarized across four dimensions:

Static authentication vulnerability. Traditional RBAC authenticates a user once, at the perimeter, and treats that authorization as valid for the remainder of the session (Sandhu et al., 1996). It provides no mechanism for continuously verifying user behavior after access has been granted.

Monolithic lateral movement. Because monolithic codebases are tightly coupled, a single compromised credential can grant an attacker broad internal reach. The absence of internal behavioral monitoring means this kind of movement frequently goes undetected by conventional, perimeter-oriented security alerts (Pereira-Vale et al., 2021).

Absence of behavioral intelligence. Most existing access-control frameworks lack AI-integrated anomaly detection, leaving organizations unable to convert large volumes of activity-log data into real-time trust assessments capable of identifying credential theft or irregular transactional patterns.

Reactive security posture. Current security models are predominantly reactive, often responding only after a breach has been confirmed. There remains a limited body of unified approaches that combine Zero-Trust principles (Rose et al., 2020) with AI-driven analysis to intercept threats proactively.

1.3. Motivation of the Study

The motivation for this study arises, first, from the practical need to modernize the security posture of monolithic architectures, which remain in wide use despite the broader industry shift toward microservices and continue to rely on outdated, static permission models. Traditional RBAC assumes a user is trustworthy once the initial authentication boundary has been passed (Sandhu et al., 1996); within a tightly coupled monolithic environment, this binary trust assumption creates a structural blind spot that can allow credential theft or insider misuse to persist undetected.

This research is further motivated by the founding premise of Zero-Trust security, namely that no user, device, or session should be implicitly trusted simply because of its location inside the network perimeter (Kindervag, 2010; Rose et al., 2020). Applying this “never trust, always verify” philosophy internally, rather than only at the network boundary, offers one route toward closing the lateral-movement gap described in Section 1.2.

The study is also motivated by the underused potential of behavioral log data. Organizations generate substantial volumes of request and access logs, but converting that raw data into a proactive defense signal remains uncommon in practice. By applying unsupervised learning specifically the Isolation Forest algorithm (Liu et al., 2008) - this research develops an intelligent middleware layer that scores environmental risk from user context without requiring a labeled dataset of known attacks, a methodological choice that matters because labeled insider-threat data is rarely available. The broader aim is a unified framework in which permissions function as dynamic trust decisions rather than static, role-based grants, so that any meaningful deviation from normal behavior triggers an immediate, automated response.

1.4. Limitations of the Study

Despite the comprehensive intent of the proposed framework, several limitations define the boundaries of this research.

First, the study is bounded in architectural scope to monolithic web applications built with PHP, Laravel, and Python. While the underlying Zero-Trust principles are broadly transferable, the specific middleware and interception mechanisms developed here are not validated against microservices or serverless deployments, and adapting them to those architectures would likely require substantial redesign of the communication and interception layers.

Second, reliance on the Isolation Forest algorithm shapes the range of attacks the system can realistically detect. Unsupervised anomaly detection performs well against behavior that diverges sharply from an established baseline, such as a sudden spike in data throughput, but it is comparatively weak against low-and-slow attacks, in which an adversary deliberately paces malicious actions to resemble normal usage over an extended period.

Third, the behavioral baseline in this study is built from a deliberately narrow feature set - login time, request frequency, and data throughput. Other potentially useful risk signals, such as IP geolocation changes or browser and device fingerprinting, fall outside this scope and could provide a more complete view of user intent in future extensions of the framework.

Fourth, the framework is exposed to the general false-positive risk that affects most behavioral anomaly detectors. Legitimate but irregular activity, such as a spike in usage around an institutional deadline, may be misclassified as suspicious, with the practical effect of disrupting a legitimate user's session.

Finally, continuous, per-request trust scoring introduces computational overhead. Because every API call must be intercepted and evaluated, the goal of seamless real-time enforcement has to be balanced against the added latency this introduces under high-traffic conditions.

1.5. Objective of the Study

The primary objective of this study is to transform the static security model of monolithic architectures into a proactive, dynamic defense mechanism by integrating AI-driven behavioral anomaly detection with Zero-Trust RBAC. In monolithic environments, where internal lateral movement often goes unrestricted after initial authentication, this research seeks to implement a continuous “**never trust, always verify**” approach to access control (Kindervag, 2010; Rose et al., 2020).

The specific objectives of the study are as follows:

1. To identify and extract behavioral features from user activity logs - including login hours, request frequency per minute, and data throughput - sufficient to establish a baseline of normal user behavior.
2. To develop an unsupervised machine learning model using the Isolation Forest algorithm (Liu et al., 2008) capable of detecting deviations associated with credential theft or insider threats without requiring labeled malicious datasets.
3. To design a Zero-Trust middleware interceptor for monolithic web applications that inspects every API request and computes a dynamic trust score in real time.
4. To establish a unified policy decision engine that synchronizes AI-generated trust assessments with RBAC permissions and automatically triggers security responses such as access restriction or session revocation.
5. To validate the effectiveness of the proposed framework through experimental testing on real-world or benchmark datasets, assessing whether the approach improves resilience against threats that evade conventional, perimeter-based defenses.

1.6. Scope of the Study

The scope of this research is defined by the following areas:

1. Architecture focus: security implementation in monolithic web environments built with PHP, Laravel, and Python.
2. Zero-Trust integration: development of an interception middleware that enforces continuous verification based on Zero-Trust principles (Rose et al., 2020).
3. AI-driven detection: application of the Isolation Forest algorithm (Liu et al., 2008) to analyze behavioral features such as login hours, request frequency, and data throughput.
4. Dynamic access control: creation of a unified policy engine that uses trust scores to dynamically enforce RBAC permissions.

This study does not address non-monolithic architectures such as microservices or serverless systems. It also excludes external network threats such as distributed denial-of-service (DDoS) attacks, hardware-level authentication mechanisms such as biometrics, and non-technical user-context factors such as psychological or socio-economic variables. In addition, the primary detection framework does not pursue supervised learning methods, owing to the general scarcity of sufficiently labeled malicious datasets for this problem domain.

1.7. Overview of Methodology

The methodology adopted in this study follows a multi-stage process. First, a monolithic web application is developed using PHP, Laravel, and Python as the implementation environment for the proposed framework. User activity data are then collected, preprocessed, and normalized using Min-Max scaling, which brings behavioral features into a consistent range between 0 and 1 before they are passed to the detection model.

During the detection phase, the Isolation Forest algorithm (Liu et al., 2008) is trained on unlabeled behavioral data to analyze key features such as login hours, request

frequency, and data throughput. Based on this analysis, the system generates real-time risk scores and corresponding trust scores for active sessions.

These scores are then enforced through a Zero-Trust middleware layer that intercepts API requests and performs continuous authorization checks, consistent with the architecture described by Rose et al. (2020). Finally, a unified policy evaluation process determines whether a session should be classified as normal, suspicious, or blocked, according to its dynamically computed trust score.

1.8. Organization of the Thesis

This thesis is organized into five chapters, each addressing a specific aspect of the research in a clear and logical sequence.

Chapter 1: Introduction presents the background, problem context, motivation, objectives, scope, limitations, and an overview of the proposed methodology.

Chapter 2: Literature Review examines existing studies related to Zero-Trust Architecture, RBAC enhancement, and machine learning techniques for behavioral anomaly detection, while identifying the research gaps addressed by this study.

Chapter 3: Methodology and Implementation explains the research design and system development framework, including the middleware interception layer, feature engineering process, Isolation Forest implementation, and the mathematical models used for normalization and trust-score computation.

Chapter 4: Experimental Results and Analysis presents the experimental setup, performance evaluation, anomaly detection results, mitigation effectiveness, and system overhead analysis under varying traffic conditions.

Chapter 5: Conclusion and Future Work summarizes the main findings, discusses the contributions and limitations of the research, and offers recommendations for future improvement and extension of the framework.

Chapter 2

Literature Review

2.1 Introduction

The rapid expansion of cloud-based and web-based applications has significantly increased security risks in modern computing environments. Traditional security models are no longer sufficient to protect systems from insider threats, credential theft, and session hijacking. As a result, Zero Trust Architecture (ZTA) has emerged as a modern security paradigm that eliminates implicit trust and continuously verifies every access request.

This chapter reviews foundational studies in Zero Trust security, machine learning-based anomaly detection, Isolation Forest-based behavioral analytics, and dynamic access control systems. It also identifies limitations in existing approaches and establishes the research gap that motivates this study. The review is structured into six main sections: foundational studies on access control models, computational approaches for behavioral anomaly detection, related studies on Zero Trust and anomaly detection, critical analysis of existing research, research gap identification, and problem statement formulation.

2.2 Foundational Studies on Access Control and Zero Trust

Traditional access control systems have formed the foundation of information security for several decades. Among these systems, Discretionary Access Control (DAC), Mandatory Access Control (MAC), and Role-Based Access Control (RBAC) have been widely implemented in enterprise environments. RBAC stands as the most commonly deployed model due to its operational simplicity, administrative scalability, and alignment with organizational hierarchies (Sandhu et al., 1996).

However, RBAC operates on a fundamental assumption of static trust that creates significant security vulnerabilities. Once users successfully authenticate, they receive role-based permissions that remain active throughout their entire session without further verification. This architectural limitation creates a critical security gap where compromised credentials can lead to unauthorized access without any additional

validation mechanisms. An attacker who obtains legitimate credentials can exploit assigned privileges indefinitely until the session terminates or manual intervention occurs (Sandhu et al., 1996).

Zero Trust Architecture (ZTA) addresses this fundamental limitation by fundamentally redefining the trust model. Originally introduced by Forrester Research in 2010 and later standardized by National Institute of Standards and Technology (NIST) Special Publication 800-207, ZTA eliminates the concept of implicit trust based on network location or initial authentication (Kindervag, 2010). Instead, ZTA enforces continuous verification of users, devices, and contextual behavior for every access request, regardless of whether the request originates from inside or outside the organizational network perimeter (Rose et al., 2020).

The architectural evolution from traditional models to Zero Trust represents a fundamental paradigm shift in security design. Traditional RBAC follows a linear authentication-to-access flow: User authenticates, system performs role check, access is granted permanently for the session duration. In contrast, Zero Trust implements a continuous verification loop: User authenticates, system performs continuous verification, evaluates contextual risk, and makes dynamic access decisions for each request (Rose et al., 2020).

2.3 Computational Approaches for Behavioral Anomaly Detection

Machine learning has emerged as a core component of modern cybersecurity systems, enabling automated detection of abnormal user behavior by learning patterns from historical data. The application of machine learning techniques to anomaly detection has transformed security operations from rule-based static detection to intelligent adaptive systems that can identify previously unknown threats.

2.3.1 Machine Learning Approaches for Anomaly Detection

Machine learning techniques employed in cybersecurity anomaly detection can be systematically categorized into three primary categories based on their learning methodology and data requirements.

Supervised Learning Approaches: Supervised learning algorithms including Random Forest, Support Vector Machines (SVM), and Artificial Neural Networks require extensively labeled datasets containing both normal and malicious activity examples. While these methods achieve high classification accuracy when trained on sufficiently large labeled datasets, their effectiveness is fundamentally limited by the scarcity of real attack data in production environments. Security systems rarely have comprehensive labeled datasets containing diverse attack scenarios, making supervised learning impractical for many real-world applications (Bishop, 2006).

Unsupervised Learning Approaches: Unsupervised learning methods operate without requiring labeled training data, making them particularly valuable for cybersecurity applications where attack data is limited or unavailable. Techniques such as k-Means clustering, Principal Component Analysis (PCA), and Isolation Forest identify anomalies by learning the structure of normal behavior and detecting deviations from established patterns. These methods are widely deployed in security systems for detecting unknown threats and novel attack patterns that supervised methods cannot recognize (Chandola et al., 2009).

Deep Learning Approaches: Advanced deep learning models represent the state-of-the-art in complex pattern recognition for security applications. Autoencoders learn compressed representations of normal behavior and identify anomalies through reconstruction error analysis. Long Short-Term Memory (LSTM) networks excel at modeling sequential behavioral patterns and temporal dependencies in user activity. Generative Adversarial Networks (GANs) can generate synthetic anomaly data for training purposes, addressing the labeled data scarcity problem (Goodfellow et al., 2016).

2.3.2 Isolation Forest for Behavioral Anomaly Detection

Isolation Forest stands as one of the most widely utilized unsupervised learning algorithms for detecting anomalies in behavioral datasets, particularly in cybersecurity applications. The algorithm operates through a fundamentally different mechanism than traditional clustering or density-based methods. Isolation Forest isolates data points through successive random partitioning of the feature space. Normal data

points require many splits to isolate because they reside in dense regions of the feature space, whereas anomalies require fewer splits because they reside in sparse regions (Liu et al., 2008).

The mathematical foundation of Isolation Forest relies on path length analysis. For each data point, the algorithm constructs multiple isolation trees (iTrees) by randomly selecting features and split values. The anomaly score is computed based on the average path length required to isolate the data point across all trees. Shorter average path lengths indicate higher anomaly scores, as anomalies are isolated more quickly than normal points (Liu et al., 2012).

In cybersecurity applications, Isolation Forest demonstrates particular effectiveness for detecting unusual login behavior patterns, abnormal session activity sequences, and suspicious access patterns that deviate from established user baselines. The algorithm's ability to handle high-dimensional feature spaces makes it well-suited for behavioral analytics, where numerous contextual features (temporal, geospatial, device, volume) must be analyzed simultaneously (Liu et al., 2008).

Table: 2.1 Key Strengths of Isolation Forest

Strength	Security Application Benefit
No labeled dataset required	Enables deployment without prior attack data
Efficient for high-dimensional data	Handles multiple behavioral features simultaneously
Suitable for real-time detection	Supports sub-second anomaly scoring for access decisions
Robust to noise	Maintains performance with imperfect behavioral data
Scalable to large datasets	Processes extensive user activity logs efficiently

2.4 Related Studies on Zero Trust and Anomaly Detection

This section provides a comprehensive synthesis of key research contributions in Zero Trust security architecture, behavioral analytics systems, and machine learning-based anomaly detection approaches. The reviewed studies represent diverse methodologies, datasets, and application domains, providing a broad foundation for understanding the current state of research.

Table 2.2: Comparative Analysis of Related Studies on Anomaly Detection in Zero Trust Architecture

SL	Author(s)	Year	Research Focus	Method / Technique	Dataset / Environment	Key Findings	Limitation(s)
1	Liu, Ting & Zhou	2008	Isolation-based anomaly detection	Isolation Forest	Synthetic + benchmark datasets	Efficient anomaly detection without labels	Not integrated with security systems
2	Sun et al.	2016	User behavior anomaly detection	Extended Isolation Forest	Enterprise logs	Detects insider anomalies effectively	No Zero Trust access control integration
3	Sharma	2023	Continuous authentication in ZTA	ML-based authentication	User activity logs	Improves continuous security monitoring	Lacks real-time decision engine
4	Khan et al.	2023	UEBA for insider threats	Behavior analytics + ML	Insider threat datasets	High detection accuracy for insider attacks	Requires labeled data
5	Hassan et al.	2025	Zero Trust ML framework	Isolation Forest + SVM	IoT behavioral data	Explainable ML improves detection trust	High computational complexity

6	Lawal et al.	2025	Zero Trust + behavioral analytics	ML-based anomaly detection	Web + network logs	Enables adaptive access control	Limited RBAC integration
7	Chen et al.	2026	Zero Trust + graph anomaly detection	GNN + ZT framework	Enterprise network traffic	Strong detection of relational anomalies	Complex model, high overhead
8	Pokhrel et al.	2024	Distributed Zero Trust system	Federated learning + anomaly detection	IoT network systems	Secure collaborative detection	Not suitable for monolithic apps
9	Xu et al.	2022	Deep anomaly detection	Deep Isolation Forest	Benchmark datasets	Better performance than classical IF	High computational cost
10	Zhang et al.	2023	Context-aware Zero Trust analytics	Feature selection + ML	User behavior logs	Improves behavioral accuracy	No direct access control integration

2.5 Critical Analysis of Existing Research

A comprehensive and critical review of the existing literature reveals several fundamental limitations that prevent current Zero Trust and behavioral anomaly detection systems from achieving their full security potential in practical web application deployments.

Limitation 1: Static RBAC Systems Lack Continuous Behavioral Verification.

Traditional Role-Based Access Control implementations, including widely used Laravel packages such as `spatie/laravel-permission`, operate on static permission assignment that remains constant throughout user sessions. Once authentication succeeds, role-based permissions are granted without any consideration of behavioral context or risk assessment. This architectural limitation means that compromised credentials grant full role privileges indefinitely, enabling attackers to perform unauthorized actions without detection (Sandhu et al., 1996).

Limitation 2: Machine Learning Models Are Not Integrated Into Real-Time Access Control.

The majority of existing research on machine learning-based anomaly detection treats detection and authorization as separate components. Anomaly detection systems generate risk scores or flags, but these outputs do not directly influence access control decisions in real-time. Instead, security teams must manually review alerts and adjust permissions, introducing significant delays between threat detection and response (Khan et al., 2023; Lawal et al., 2025).

Limitation 3: Existing Zero Trust Implementations Focus Primarily on Network Security

Current Zero Trust Architecture deployments predominantly target network-level security controls, including micro-segmentation, network traffic monitoring, and device verification. These implementations rarely extend Zero Trust principles to application-level access control within web applications. Consequently, web applications remain vulnerable to insider threats even when network-level Zero Trust is properly implemented (NIST, 2020; Lawal et al., 2025).

2.6 Research Gap and Problem Statement

2.6.1 Research Gap Identification

Based on comprehensive literature analysis and critical evaluation of existing research, the following specific gaps are identified that this thesis addresses:

1. Lack of Integration Between RBAC and Behavioral Anomaly Detection

Existing systems implement Role-Based Access Control and behavioral anomaly detection as separate, independent components. RBAC manages static permissions based on user roles, while behavioral analytics monitors user activity for anomalies. However, no unified framework exists that integrates behavioral risk scores directly into RBAC decision-making to enable dynamic, risk-adaptive permission adjustment (Khan et al., 2023; Zhang et al., 2023).

2. Limited Application of Isolation Forest in Real-Time Access Control Systems

While Isolation Forest demonstrates strong anomaly detection performance in research settings (Liu et al., 2008; Liu et al., 2012; Xu et al., 2022), existing literature shows minimal deployment of Isolation Forest within real-time access control engines. The algorithm is typically used for post-incident detection or offline analysis rather than for immediate access restriction during active sessions (Xu et al., 2022).

3. Absence of Continuous Trust Evaluation in Monolithic Web Applications

Continuous trust evaluation, a core Zero Trust principle, is implemented in cloud-native and microservices architectures (Pokhrel et al., 2024; Lawal et al., 2025; Chen et al., 2026), but no research addresses continuous trust evaluation within monolithic web application architectures. Monolithic applications face unique constraints including centralized state management, single-server deployment, and unified logging that differ from distributed architectures (Zhang et al., 2023).

4. No Unified Framework Combining Authentication, Anomaly Detection, and Decision-Making

Current research treats authentication, anomaly detection, and authorization as separate problems addressed by different systems. No unified framework exists that integrates these three components into a cohesive security architecture where authentication initiates the session, anomaly detection continuously monitors behavior, and decision-making dynamically adjusts access based on risk (NIST, 2020; Lawal et al., 2025).

5. Lack of Lightweight, Deployable Zero Trust Systems for Real-World Applications
Advanced Zero Trust implementations often require complex infrastructure, distributed deployment, and significant computational resources (Pokhrel et al., 2024; Chen et al., 2026). This complexity prevents adoption by organizations with limited budgets, infrastructure constraints, or technical expertise. No lightweight, easily deployable Zero Trust system exists that maintains strong security while remaining practical for real-world deployment in resource-constrained environments (Lawal et al., 2025).

2.6.2 Problem Statement

Existing security systems in modern web applications rely heavily on static Role-Based Access Control mechanisms that grant access privileges after initial authentication without subsequent continuous validation throughout the session duration. This fundamental architectural limitation creates significant security vulnerabilities where attackers who obtain compromised credentials can exploit assigned privileges indefinitely, maintaining unauthorized access and performing malicious actions without detection or automatic restriction.

Although Zero Trust Architecture introduces the critical principle of continuous verification to address static trust limitations, most existing Zero Trust implementations focus primarily on network-level security controls and do not integrate behavioral anomaly detection mechanisms or real-time automated decision-making capabilities within web application access control. Additionally, machine learning-based security systems for anomaly detection are typically deployed as independent monitoring components that operate separately from authorization frameworks, limiting their practical security effectiveness because detected anomalies do not directly influence access decisions.

Furthermore, existing research on Zero Trust and behavioral analytics predominantly targets cloud-native, microservices, or distributed architectures, leaving monolithic web application architectures (such as Laravel-based applications) without tailored implementation guidance despite their continued prevalence in many organizations, particularly those with infrastructure budget constraints.

Therefore, there exists a critical need for an intelligent, integrated security framework that combines three core components: (1) Role-Based Access Control for organizational permission management, (2) Zero Trust principles for continuous verification and dynamic trust evaluation, and (3) Isolation Forest-based behavioral anomaly detection for real-time risk assessment. This unified framework must enable continuous, adaptive, and context-aware access control specifically designed for monolithic web application architectures, addressing the identified research gaps while maintaining practical deployability for real-world organizational deployment.

Chapter Three

Methodology and Implementation

3.1 Introduction

This chapter presents the methodology, system design, and experimental framework of the proposed Zero-Trust Role-Based Access Control (RBAC) system with integrated behavioral anomaly detection. The primary objective of the framework is to extend conventional RBAC by incorporating a machine learning-based behavioral verification layer that continuously evaluates user requests before permitting access to protected application resources.

Conventional RBAC systems govern access through static role-permission mappings. While this approach is widely adopted in enterprise environments due to its administrative simplicity and scalability, it carries a fundamental security limitation: once a user has been authenticated and assigned a valid role, the system implicitly trusts all subsequent actions within that role's permission boundary (Sandhu et al., 1996). This assumption creates a significant attack surface in scenarios involving compromised credentials, insider threats, or session hijacking, where a legitimate account may be exploited by a malicious actor. The Zero-Trust security paradigm directly addresses this vulnerability by adopting the principle that no request should be inherently trusted, regardless of origin or prior authentication status (Kindervag, 2010; Rose et al., 2020). Every access attempt must therefore be validated using both authorization rules and real-time contextual trust signals.

To address these limitations, this study proposes and implements a behavior-aware Zero-Trust RBAC architecture within a Laravel-based web application. The framework integrates four core components: a Laravel application for authentication, route protection, and RBAC enforcement; Redis for caching trust decisions to minimize latency; a FastAPI microservice for exposing the anomaly detection endpoint; and a Weighted Isolation Forest model for evaluating suspicious behavioral patterns at runtime. The model operates on eight engineered behavioral features extracted from login events and request activity, including request frequency, login hour, failed login attempts, IP irregularity, session duration, data transfer volume, role-route consistency, and endpoint sensitivity. These features are normalized, weighted according to their security relevance, and submitted to the anomaly detector, which returns a trust decision. Requests exhibiting anomalous behavioral signatures

are either blocked or flagged for secondary review, thereby implementing a continuous verification loop consistent with Zero-Trust principles.

The chapter is organized as follows. Section 3.2 describes the research design and the overall methodological workflow. Section 3.3 presents the proposed system architecture and its constituent components. Section 3.4 explains the behavioral feature engineering strategy. Section 3.5 describes the data sources and dataset preparation process. Section 3.6 presents the mathematical formulation of the Weighted Isolation Forest model. Section 3.7 explains the Laravel middleware implementation and the runtime access verification workflow. Section 3.8 describes the Python-based model training pipeline and FastAPI deployment. Section 3.9 introduces the baseline models used for comparative evaluation. Section 3.10 defines the experimental setup and evaluation metrics. Section 3.11 provides a summary of the chapter.

3.2 Research Design

This study adopts a quantitative experimental research design combined with a prototype-based implementation methodology. The objective is not merely to propose a conceptual security model but to construct a working system prototype that can be empirically evaluated with respect to anomaly detection performance and access control effectiveness. This dual orientation theoretical framing followed by empirical validation is consistent with design science research traditions in information systems and cybersecurity (Hevner et al., 2004).

The research process is structured across five sequential phases, each building upon the output of the preceding stage.

Phase 1: Framework Design

A Zero-Trust RBAC architecture was designed by combining role-permission enforcement with a machine learning-based behavioral verification layer. The framework was implemented as a Laravel web application integrated with a Python

anomaly detection microservice. Design decisions were informed by current literature on Zero-Trust architectures, User and Entity Behavior Analytics (UEBA), and unsupervised anomaly detection.

Phase 2: Data Collection and Behavioral Logging

User interactions, login events, protected route access patterns, and session-level request data were systematically collected from the Laravel application. An external anomaly-oriented reference dataset was additionally incorporated to augment the behavioral modeling pipeline with a broader representation of irregular access patterns that may not be sufficiently present in application-specific logs alone.

Phase 3: Feature Engineering and Dataset Preparation

Raw application logs were transformed into a structured behavioral feature representation suitable for unsupervised anomaly detection. Eight features were engineered to capture distinct categories of suspicious behavior, including access timing, request bursting, credential misuse, network irregularity, and privilege inconsistency. Both application logs and external data records were mapped into this unified feature space prior to model training.

Phase 4: Model Training and Deployment

A Weighted Isolation Forest model was trained on the prepared behavioral dataset. The trained model, normalization scaler, and feature weight vector were serialized and deployed through a FastAPI microservice, enabling real-time inference during Laravel request processing. Baseline models were also trained under the same conditions to support comparative analysis.

Phase 5: Comparative Evaluation

The proposed Weighted Isolation Forest model was evaluated against two baseline anomaly detection methods: Standard Isolation Forest and Local Outlier Factor (LOF). All models were assessed using the same eight-feature representation, the same train-test partition, and the same evaluation metrics, including precision, recall, F1-score, and confusion matrix analysis. These metrics are standard in anomaly detection research because accuracy alone can be misleading when class distributions are imbalanced (Chandola et al., 2009).

3.3 Proposed System Architecture

The proposed framework constitutes a hybrid access-control architecture that extends traditional RBAC with a machine learning-based anomaly detection layer. Rather than trusting an authenticated request solely on the basis of role membership, the system evaluates whether the behavioral signature of each request is consistent with established normal usage patterns. This approach embodies the Zero-Trust principle of continuous, context-aware verification (NIST, 2020; Rose et al., 2020). The architecture is composed of five interdependent components, described in the subsections that follow.

3.3.1 User Interaction Layer

This layer represents authenticated users interacting with the web application through various protected operations, including login, dashboard navigation, task management, profile updates, and administrative functions. All user-initiated requests must traverse the subsequent authentication and authorization layers before access is granted.

3.3.2 RBAC and Middleware Layer

Laravel handles authentication and role-based authorization using configured role-permission mappings. Upon successful RBAC validation, a custom middleware intercepts the request to extract behavioral signals from the current request context and recent activity history. These signals are assembled into a structured feature vector that serves as the input to the anomaly detection pipeline.

3.3.3 Cache and Queue Layer

Redis is employed to store recent anomaly decisions using a hash-based cache key derived from the user identifier and behavioral feature vector. When a request produces a feature signature that matches a recently cached decision, the system reuses the cached result, eliminating the need for a new model inference call. This strategy significantly reduces latency and ensures the Zero-Trust verification overhead remains acceptable for web application deployment.

3.3.4 Machine Learning Inference Layer

A Python-based FastAPI microservice hosts the trained Weighted Isolation Forest model along with the normalization scaler and weight vector. Upon receiving a behavioral feature vector from the Laravel middleware, the service applies the normalization and weighting transformations before invoking the model to generate an anomaly label and anomaly score.

3.3.5 Decision Layer

Based on the anomaly score returned by the model and the configured decision threshold, the system enforces one of three outcomes: the request is permitted if behavioral patterns are consistent with normal activity; the request is flagged for secondary review if the score indicates moderate risk; or the request is blocked and an access-denied response is returned if the model identifies strongly anomalous behavior.

This layered design is consistent with risk-based access validation approaches described in recent Zero-Trust and UEBA frameworks, wherein session-level behavioral features are passed to anomaly detection components and mapped to tiered access decisions (Wijesekara & Rupasinghe, 2023; Xiao et al., 2022).

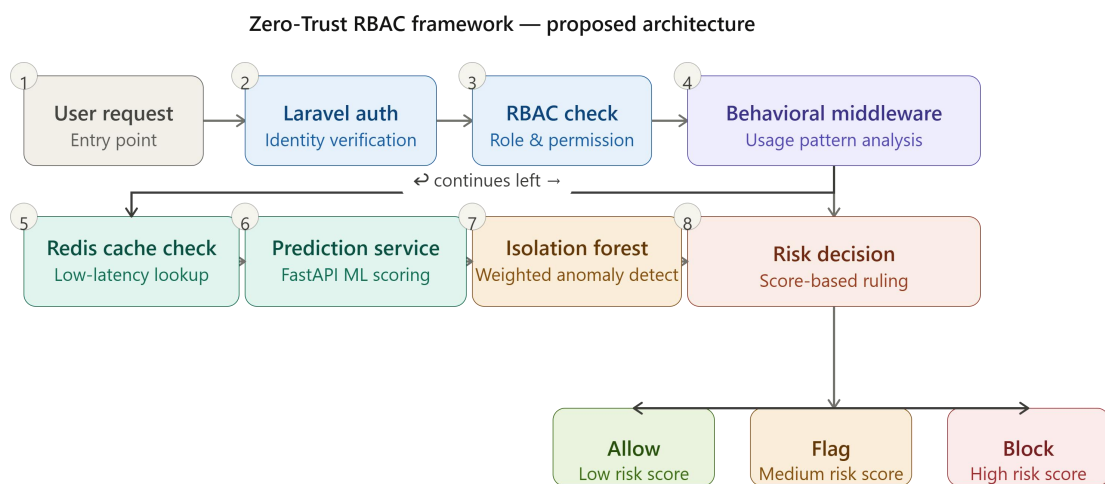


Figure 3.1. Proposed Zero-Trust RBAC framework integrating Laravel middleware, Redis caching, FastAPI inference, and Weighted Isolation Forest anomaly detection.

Table 3.1 Major Components of the Proposed Framework

Component	Technology	Purpose
Web Application	Laravel	Core application framework
Authentication	Laravel Auth	User login and session management
RBAC Enforcement	Spatie / Custom RBAC	Role-permission authorization
Behavioral Middleware	Custom Laravel Middleware	Extract request-level behavioral features
Cache Layer	Redis	Cache recent anomaly decisions
ML API Service	FastAPI	Expose anomaly prediction endpoint
Detection Model	Weighted Isolation Forest	Predict anomalous user behavior
Database	MySQL	Store users, permissions, logs, and tasks

3.4 Behavioral Feature Engineering

The central contribution of the proposed framework lies in the use of behavioral features to supplement conventional RBAC access decisions. Rather than granting access solely on the basis of valid role membership, the system evaluates whether a given request exhibits behavioral characteristics consistent with legitimate usage. Feature engineering is therefore a critical step in the pipeline, as the anomaly detection model does not inherently understand abstract concepts such as privilege misuse or credential abuse; instead, it learns from measurable behavioral signals expressed as numerical feature values (Chandola et al., 2009).

Contemporary Zero-Trust and UEBA-oriented anomaly detection research similarly relies on session-level behavioral attributes, including login timing, failed authentication counts, privilege events, and data access volumes, to construct representations of risky user activity (Wijesekara & Rupasinghe, 2023; Liu et al., 2019). The present study operationalizes this principle through eight engineered behavioral features, each described in the following subsections.

3.4.1 Login Hour (login_hour)

This feature captures the hour of the day at which the login or access request occurs, represented as an integer in the range 0 to 23. Unusual activity occurring outside an organization's standard operating hours, such as requests made in the early morning or during weekends, may constitute a temporal anomaly indicative of unauthorized access or account compromise (Buczak & Guven, 2016).

3.4.2 Requests per Minute (req_per_min)

This feature measures the intensity of recent request activity within a rolling one-minute time window. Elevated request frequencies may indicate automated tooling, web scraping, brute-force enumeration, or application-layer denial-of-service behavior. Unusually high request rates are commonly associated with non-human or malicious actors (Scarfone & Mell, 2007).

3.4.3 IP Entropy (ip_entropy)

IP entropy quantifies the degree of irregularity in the network addresses associated with a user's recent sessions. Frequent IP address changes, or activity from geographically or topologically inconsistent addresses, may indicate the use of anonymizing proxies, virtual private networks, or unauthorized remote access. This feature captures network-level behavioral variation as a scalar irregularity score.

3.4.4 Session Duration (session_min)

Session duration, expressed in minutes, describes the elapsed time of the user's active application session. Abnormally short sessions may suggest automated interaction or premature termination, while excessively long sessions may indicate session persistence abuse or idle session exploitation. Both extremes represent deviations from typical human usage patterns.

3.4.5 Failed Login Attempts (failed_logins)

1. This feature counts the number of failed authentication attempts associated with a user account or email address within a recent time window. Elevated counts of failed logins are a well-established indicator of password guessing, credential stuffing, or brute-force attack behavior (Scarfone & Mell, 2007). This feature provides a direct signal of authentication-layer threat activity.

3.4.6 Data Transfer Volume (data_kb)

Data transfer volume approximates the quantity of data transferred or requested during the current interaction, expressed in kilobytes. Anomalously large data movements relative to a user's typical behavior may indicate data exfiltration attempts, unauthorized bulk downloads, or scraping activity (Liu et al., 2019). This feature captures the data-layer risk dimension of user behavior.

3.4.7 Role Mismatch (role_mismatch)

Role mismatch is a binary feature that indicates whether the route accessed by the user is inconsistent with their assigned role. A value of one is assigned when a user attempts to access a route categorized as belonging to a privilege level higher than their role assignment; a value of zero indicates role-route consistency. This feature directly operationalizes the concept of privilege misuse within the RBAC context.

3.4.8 Endpoint Risk (endpoint_risk)

Endpoint risk is a continuous scalar that represents the sensitivity level of the requested application route, assigned through a predefined risk mapping. Administrative, export, deletion, and user-management endpoints receive higher risk scores, while lower-sensitivity routes such as dashboards or profile pages receive lower values. This feature ensures that access attempts to high-privilege endpoints receive proportionally greater scrutiny during anomaly evaluation.

Table 3.2. Extracted behavioral features for user anomaly detection and risk scoring.

Feature	Description	Security Relevance
login_hour	Time of login or request	Detect unusual access timing
req_per_min	Recent request frequency	Detect burst traffic and automation
ip_entropy	IP irregularity score	Detect suspicious network behavior
session_min	Session duration in minutes	Detect abnormal session patterns
failed_logins	Count of recent failed logins	Detect credential-based attacks
data_kb	Approximate data transferred or requested	Detect excessive data access or exfiltration
role_mismatch	Role-route inconsistency binary flag	Detect privilege misuse attempts
endpoint_risk	Sensitivity score of requested route	Prioritize high-risk access attempts

3.4.9 Feature Weighting Strategy

Given that behavioral signals differ in their security relevance, a feature weighting mechanism is applied after normalization to amplify the influence of high-risk attributes during anomaly scoring. For instance, features such as failed login counts, role-route mismatch, and request burst frequency carry greater diagnostic value for security anomaly detection than access hour alone. The proposed model therefore incorporates a predefined weight vector W , applied element-wise to the normalized feature matrix prior to both training and inference. This approach is referred to as a Weighted Isolation Forest, as the weighting is applied at the feature representation level without altering the internal partitioning logic of the Isolation Forest algorithm.

The representative weight vector used in this study is defined as follows:

$$W = [0.9, 1.4, 1.5, 0.8, 1.3, 0.7, 1.6, 1.2]$$

where the elements correspond respectively to the features: [login_hour, req_per_min, ip_entropy, session_min, failed_logins, data_kb, role_mismatch, endpoint_risk]. These weights may be calibrated empirically based on validation performance, security domain expertise, or application-specific threat modeling requirements.

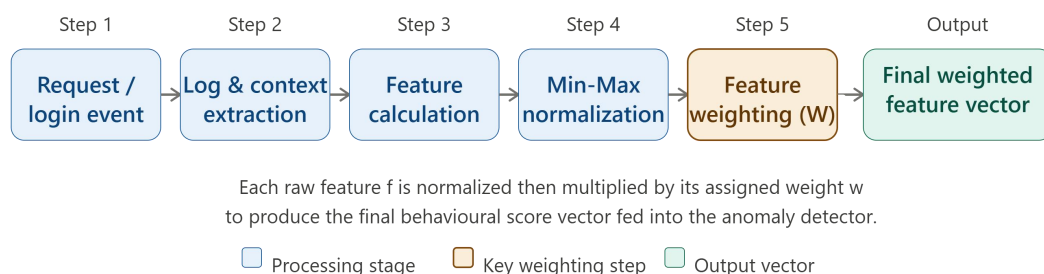


Figure 3.2. Behavioral feature extraction pipeline: login events parsed, normalized, and weighted to produce the final feature vector.

3.5 Data Sources and Dataset Preparation

The anomaly detection model requires a structured behavioral dataset that captures both normal and abnormal patterns of user activity within the application environment. To construct this dataset, the study draws from two complementary data sources: application-specific behavioral logs collected from the Laravel prototype, and an external anomaly reference dataset incorporated to broaden the model's exposure to irregular behavioral patterns. This dual-source strategy is consistent with recent

anomaly detection research, which commonly combines application-generated logs with supplementary datasets to address data sparsity in the abnormal class (Chandola et al., 2009).

3.5.1 Laravel Application Logs

The Laravel application serves as the primary behavioral data environment. Event logs are collected from user authentication processes, protected route access operations, task creation and modification events, failed login attempts, session timestamps, IP address metadata, role-based access patterns, and per-user request counts within rolling time windows. These logs directly reflect authentic user interaction with the RBAC-protected system and therefore provide a realistic and application-relevant foundation for the anomaly detection model.

3.5.2 External Anomaly Reference Dataset

An external anomaly-related dataset is incorporated to enrich the training pipeline with a broader range of abnormal behavioral patterns that may not be sufficiently represented within a single prototype application's activity logs. The purpose of this dataset is supplementary rather than substitutive: it provides additional anomalous behavior examples that expand the model's exposure during training. Where the raw structure of the external data differs from the eight-feature web request schema defined in Section 3.4, appropriate feature mapping and transformation procedures are applied to produce a unified behavioral representation prior to model training.

3.5.3 Dataset Preparation Workflow

The full dataset preparation pipeline consists of six sequential processing stages: (a) log collection from Laravel authentication and request activity; (b) data cleaning to remove duplicate, incomplete, or structurally inconsistent records; (c) feature mapping to compute the eight behavioral features from raw log fields; (d) Min-Max normalization to scale numerical features to a comparable range; (e) feature weighting by element-wise multiplication with the weight vector W ; and (f) train-test partitioning for model evaluation. Figure 3.3 depicts this preparation pipeline.

Figure 3.3. Dataset Preparation Pipeline

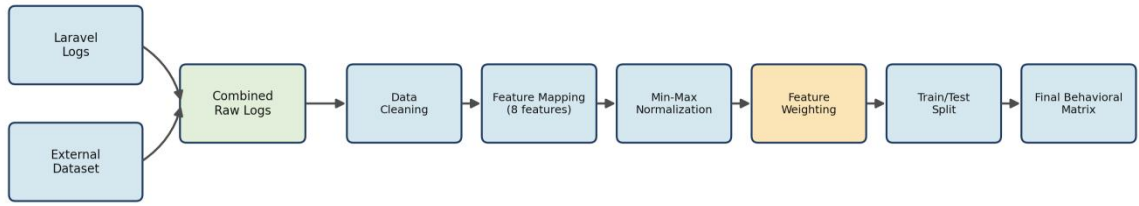


Figure 3.3. End-to-end data preparation workflow for constructing the behavioral anomaly detection dataset from Laravel application logs and external anomaly references.

3.6 Weighted Isolation Forest Model Formulation

The anomaly detection component of the proposed framework is built upon the Isolation Forest algorithm, an unsupervised ensemble method that identifies anomalous observations by measuring the ease with which they can be isolated from the remainder of the dataset through recursive random binary partitioning (Liu et al., 2008). The fundamental intuition of Isolation Forest is that anomalous data points, being fewer in number and more distant from the data bulk, require fewer random partitions to isolate than normal observations. This property makes the algorithm computationally efficient and well-suited for high-dimensional tabular data without requiring labeled attack instances.

Isolation Forest is selected for this study for several reasons. First, it operates effectively in an unsupervised regime, which is essential given the scarcity of ground-truth anomaly labels in web application access data. Second, its computational complexity is substantially lower than density-based or deep learning alternatives, making it viable for near-real-time inference in a web application context. Third, it has been successfully applied in recent cybersecurity and Zero-Trust anomaly detection research (Erfani et al., 2016; Xiao et al., 2022). The proposed enhancement introduces a feature-level weighting mechanism that biases the isolation process toward security-relevant behavioral attributes.

3.6.1 Feature Normalization

Let the raw behavioral feature matrix extracted from the dataset be defined as:

$$X = [x_1, x_2, \dots, x_n]$$

Because the individual features operate on different numeric scales, Min-Max normalization is applied to rescale all feature dimensions to the range $[0, 1]$ prior to model training:

$$X_{\text{norm}} = \text{Normalize}(X) = (X - X_{\text{min}}) / (X_{\text{max}} - X_{\text{min}})$$

This normalization ensures that no single feature dominates the partitioning process due to differences in numeric magnitude.

3.6.2 Feature Weighting

Following normalization, a predefined weight vector W is applied to the normalized feature matrix through element-wise multiplication, yielding the weighted feature representation:

$$X_w = X \odot W = [X_1W_1, X_2W_2, \dots, X_8W_8]$$

where $\odot$ denotes the Hadamard (element-wise) product. This operation increases the effective variance of security-critical features relative to lower-weight attributes, thereby directing the Isolation Forest's random partitioning process toward the most behaviorally significant dimensions.

3.6.3 Weighted Isolation Forest Training

The Weighted Isolation Forest does not alter the internal tree construction logic of the standard Isolation Forest algorithm. Instead, it modifies the input feature representation so that the forest is trained on a security-weighted behavioral space. Formally, the weighted training matrix is defined as:

$$X_w = X_{\text{norm}} \odot W$$

The Isolation Forest ensemble is then fitted to X_w . During runtime inference, the same normalization scaler and weight vector are applied to incoming request feature vectors before prediction, ensuring consistency between training and deployment representations.

Algorithm : Trust Score Prediction and Access Decision

Input : User behavioral session X

Output: Trust score T , Classification C

- 1: Receive behavioral features X
- 2: Load trained model M
- 3: Load scaler S
- 4: Load weight vector W
- 5: Normalize X using S
- 6: Apply feature weights:

$$X_w = X_{\text{norm}} \odot W$$
- 7: Compute anomaly score A using M
- 8: Convert anomaly score to trust score T
- 9: if $T \leq 0.3$ then
- 10: C $\leftarrow$ BLOCKED
- 11: else if $T \leq 0.6$ then
- 12: C $\leftarrow$ SUSPICIOUS
- 13: else
- 14: C $\leftarrow$ NORMAL
- 15: end if
- 16: Return T, C

3.7 Laravel Middleware and Runtime Access Verification

This section describes how the trained anomaly detection model is integrated into the Laravel application at runtime through a custom middleware component. The middleware ensures that each request to a protected route undergoes both RBAC authorization and behavioral anomaly evaluation before access is granted, thereby implementing a continuous verification loop consistent with the Zero-Trust security model.

3.7.1 Runtime Verification Workflow

When an authenticated user initiates a request to a protected route, the system executes the following verification sequence:

1. Authenticate the user via Laravel session or API token.

2. Validate RBAC permissions for the requested route.
3. Extract behavioral features from the request context and recent activity logs.
4. Query the Redis cache for a recent anomaly decision matching the feature signature.
5. If no cached result exists, transmit the feature vector to the FastAPI inference endpoint.
6. Receive the anomaly label and score from the Weighted Isolation Forest model.
7. Apply the risk decision: allow, flag, or block the request based on the anomaly outcome.
8. Log the decision and cache the result for subsequent reuse.

3.7.2 Middleware Implementation

The following PHP implementation illustrates the `WeightedZeroTrustMiddleware` class, which orchestrates behavioral feature extraction, Redis cache interaction, and FastAPI inference within the Laravel request lifecycle.

```
<?php
namespace App\Http\Middleware;
use Closure;
use Illuminate\Http\Request;
use Illuminate\Support\Facades\Cache;
use Illuminate\Support\Facades\Http;
use Symfony\Component\HttpFoundation\Response;

class WeightedZeroTrustMiddleware
{
    public function handle(Request $request, Closure $next): Response
    {
        $user = auth()->user();
        // Reject unauthenticated users
        if (!$user) {
```

```

return response()->json([
  'message' => 'Unauthorized'
], 401);
}

// Extract behavioral and contextual features
$features = [
  'login_hour' => now()->hour,
  'req_per_min' => $this->getRequestRate($user->id),
  'ip_entropy' => $this->calculateIpEntropy($user->id),
  'session_min' => $this->getSessionDuration($request),
  'failed_logins' => $this->getFailedLogins($user->email),
  'data_kb' => round(strlen($request->getContent()) / 1024, 2),
  'role_mismatch' => $this->checkRoleMismatch($user, $request),
  'endpoint_risk' => $this->getEndpointRisk($request->path()),
];

// Create cache key for previously computed risk
$cacheKey = 'risk:' . $user->id . ':' . md5(json_encode($features));

// Retrieve risk score from cache or call ML service
if (Cache::has($cacheKey)) {
  $risk = Cache::get($cacheKey);
} else {
  $response = Http::post(
    'http://127.0.0.1:8001/predict',
    $features
  );
  $risk = $response->json();
  Cache::put($cacheKey, $risk, now()->addMinutes(10));
}

// Block request if anomaly is detected
if (($risk['is_anomaly'] ?? false) === true) {
  return response()->json([
    'message' => 'Suspicious activity detected. Access blocked.'
  ], 403);
}

```

```

}
// Allow request to proceed
return $next($request);
}
}
?>

```

3.7.3 Feature Extraction Helper Functions

The middleware relies on private helper methods to derive behavioral feature values from application logs and request metadata. The following implementations illustrate the core extraction logic for request rate, failed login count, endpoint risk classification, and role-route mismatch detection.

3.7.4 Redis Caching Strategy

To minimize the performance overhead associated with repeated model inference calls, Redis is used to cache anomaly decisions keyed by a combination of the user identifier and an MD5 hash of the behavioral feature vector. When a subsequent request produces an identical or closely matching feature signature within the cache validity window of ten minutes, the previously computed anomaly decision is retrieved directly from the cache rather than invoking a new FastAPI prediction. This caching strategy maintains the responsiveness of the web application while preserving the continuous verification property of the Zero-Trust framework. Figure 3.4 depicts the complete runtime request verification flow.

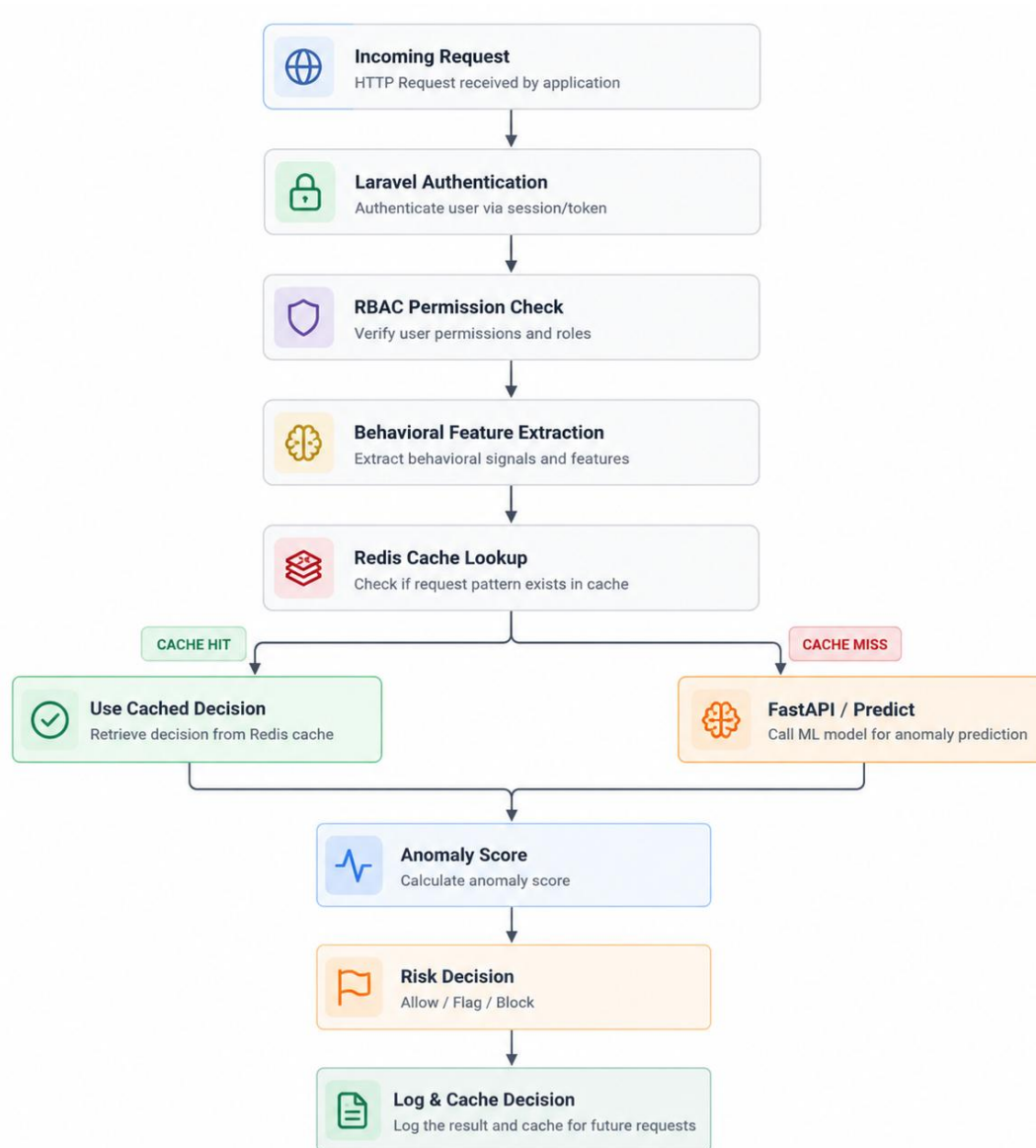


Figure 3.4. Runtime flow of request validation

3.8 Python Model Training and FastAPI Deployment

This section describes the Python-based training pipeline used to produce the Weighted Isolation Forest model and its deployment as a FastAPI microservice accessible to the Laravel application at runtime. The training implementation uses the pandas, NumPy, scikit-learn, and joblib libraries.

3.8.1 Dataset Loading and Feature Selection

The behavioral dataset is loaded from a structured CSV file and the eight engineered features are extracted as the model input matrix. The following code segment demonstrates this initialization step.

3.8.2 One-Class SVM Training (Baseline Model)

As a complementary baseline, a One-Class Support Vector Machine (OCSVM) is trained using the same normalized behavioral feature matrix without feature weighting. The model employs a Radial Basis Function (RBF) kernel with gamma set to "scale" and ν (nu) = 0.05, where the ν parameter specifies the expected upper bound on the proportion of anomalous samples within the training data. The OCSVM learns a decision boundary that encloses the majority of normal behavioral observations, classifying samples falling outside this boundary as anomalies. Following training, the fitted model and the corresponding normalization scaler are serialized using joblib to support deployment within the behavioral access control framework. The One-Class SVM serves as a representative boundary-based anomaly detection algorithm, providing a meaningful comparison against both the proposed Weighted Isolation Forest and the density-based Local Outlier Factor (Schölkopf et al., 2001).

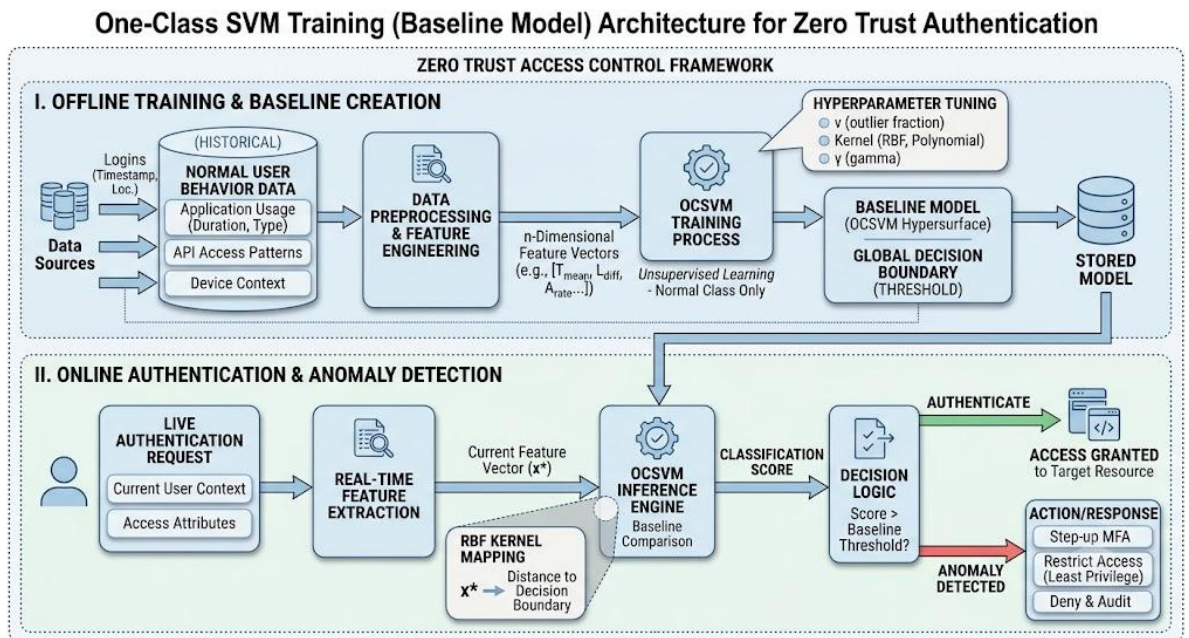


Figure 3.5: One-Class SVM Architecture for Zero Trust Authentication

Here is a breakdown of how the architecture works, followed by the core algorithm and its operational steps:

1. Data Preparation

Step 1: Capture user logs (Time, IP, Device, Behavior).

Step 2: Convert text data into numbers (e.g., "Monday"-->1).

Step 3: Group these numbers into a single coordinate point (vector).

Step 4: Scale the numbers so large metrics don't skew the model.

2. Training (Baseline)

Step 5: Collect 30 days of verified, "clean" user behavior data.

Step 6: Pick the RBF kernel to handle complex, non-linear behavior shapes.

Step 7: Set ν (the percentage of acceptable false alarms in training).

Step 8: Run the algorithm to draw a tight boundary around the normal data.

Step 9: Save the mathematical boundary edge (Support Vectors) as the user's profile.

3. Live Check (Inference)

Step 10: Intercept a new, live user access request.

Step 11: Format and scale this live request into a coordinate point.

Step 12: Fetch the user's saved baseline profile.

Step 13: Check if the new point falls inside (+1) or outside (-1) the boundary.

4. Zero Trust Action

Step 14: IF +1: Smoothly grant access.

Step 15: IF -1 (Close to edge): Prompt for MFA (Step-up).

Step 16: IF -1 (Far from edge): Block access and kill the session.

3.8.3 Local Outlier Factor (LOF) Model

Local Outlier Factor (LOF) is an unsupervised anomaly detection technique that identifies unusual data instances through local density comparison. Instead of relying on predefined labels, the method evaluates how densely a data point is surrounded by neighboring observations and determines whether it significantly deviates from its local environment.

The LOF approach was developed to address limitations associated with global anomaly detection methods. In many real-world datasets, certain observations may appear normal when viewed across the entire dataset but exhibit abnormal characteristics when examined within their immediate neighborhood. LOF is specifically designed to capture such localized irregularities.

Unlike supervised learning algorithms that depend on annotated training data, LOF operates independently of class labels. It estimates the degree of abnormality by measuring the relative isolation of each observation compared with nearby data points. As a result, LOF is particularly useful in situations where labeled anomaly data is unavailable or difficult to obtain.

The underlying principle of LOF is that observations located in regions of substantially lower density than surrounding regions are more likely to represent anomalies. This localized density-based analysis enables the model to detect subtle behavioral deviations that may not be identified through conventional distance-based approaches.

Because of these characteristics, LOF has been applied across multiple domains requiring abnormal behavior detection and continuous monitoring.

Common application areas include:

- Zero Trust Authentication frameworks
- Intrusion Detection Systems (IDS)
- Fraud and financial anomaly detection
- Network traffic and cybersecurity monitoring

- Healthcare data analysis and patient monitoring
- Risk assessment and financial analytics

LOF remains a widely used density-based anomaly detection technique because of its ability to discover localized outliers while operating effectively without labeled datasets.

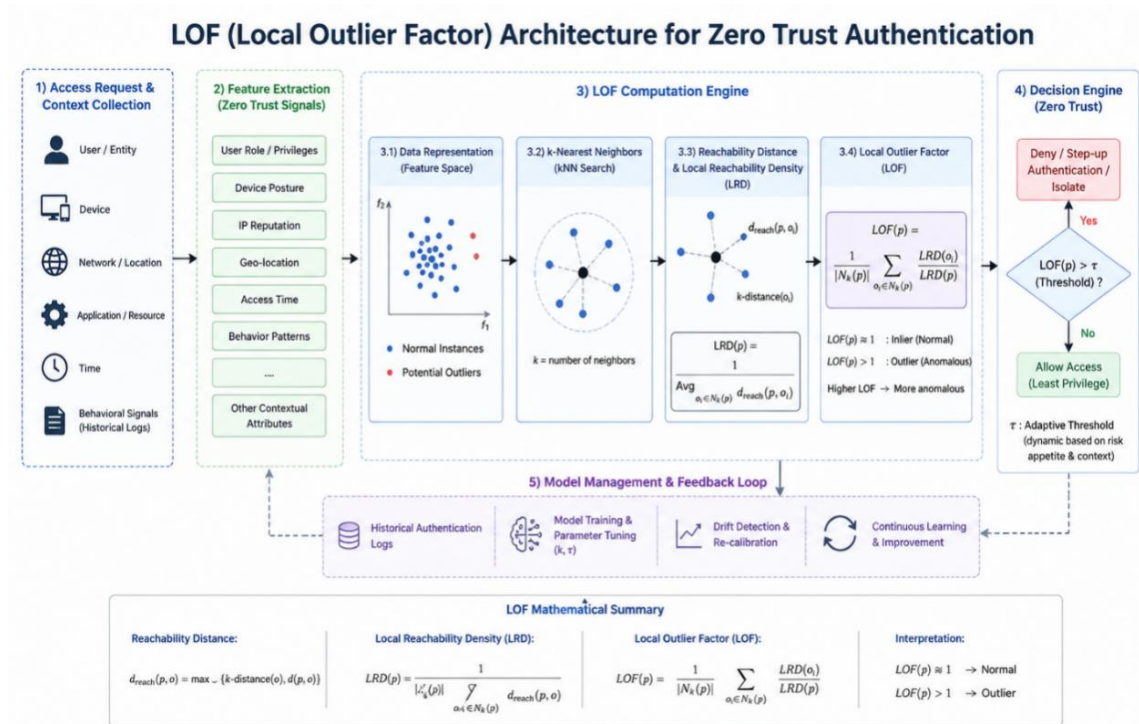


Figure: LOF Architecture for Zero Trust Authentication

Figure 3.6: LOF Architecture for Zero Trust Authentication

Algorithm of Local Outlier Factor (LOF):

Input: Dataset (D), Number of Neighbors (K)

Step 1: Data Preparation

Prepare and preprocess the dataset by cleaning missing values and normalizing features.

Step 2: Calculate Distance

Compute the distance between each observation and all other observations.

Step 3: Find K-Nearest Neighbors

Identify the nearest (K) neighboring points for each observation.

Step 4: Calculate Reachability Distance

Compute the reachability distance to reduce the effect of extremely close points.

Step 5: Compute Local Reachability Density (LRD)

Calculate local density using reachability distances.

Step 6: Calculate LOF Score

Compare local density with neighboring densities.

Step 7: Classification

Assign labels based on LOF score:

$\text{LOF} \approx 1 \rightarrow \text{Normal}$

$\text{LOF} > 1 \rightarrow \text{Outlier}$

Higher LOF $\rightarrow$ Stronger anomaly

Output: Normal (0) or Anomaly (1)

3.8.4 Weighted Isolation Forest Model

Weighted Isolation Forest (WIF) is an enhanced version of the traditional Isolation Forest algorithm developed for anomaly detection. It improves anomaly identification by assigning weights to observations or isolation paths instead of treating all partitions equally.

The fundamental principle of Weighted Isolation Forest is that anomalous observations are isolated more quickly than normal observations during random partitioning. The weighting mechanism further strengthens the contribution of important observations and improves detection accuracy.

Compared with standard Isolation Forest, WIF can provide better performance on imbalanced datasets and complex data distributions.

Common application areas include:

- Zero Trust Authentication
- Intrusion Detection Systems (IDS)
- Fraud Detection
- Network Security Monitoring
- Financial Risk Analysis
- Healthcare Anomaly Detection

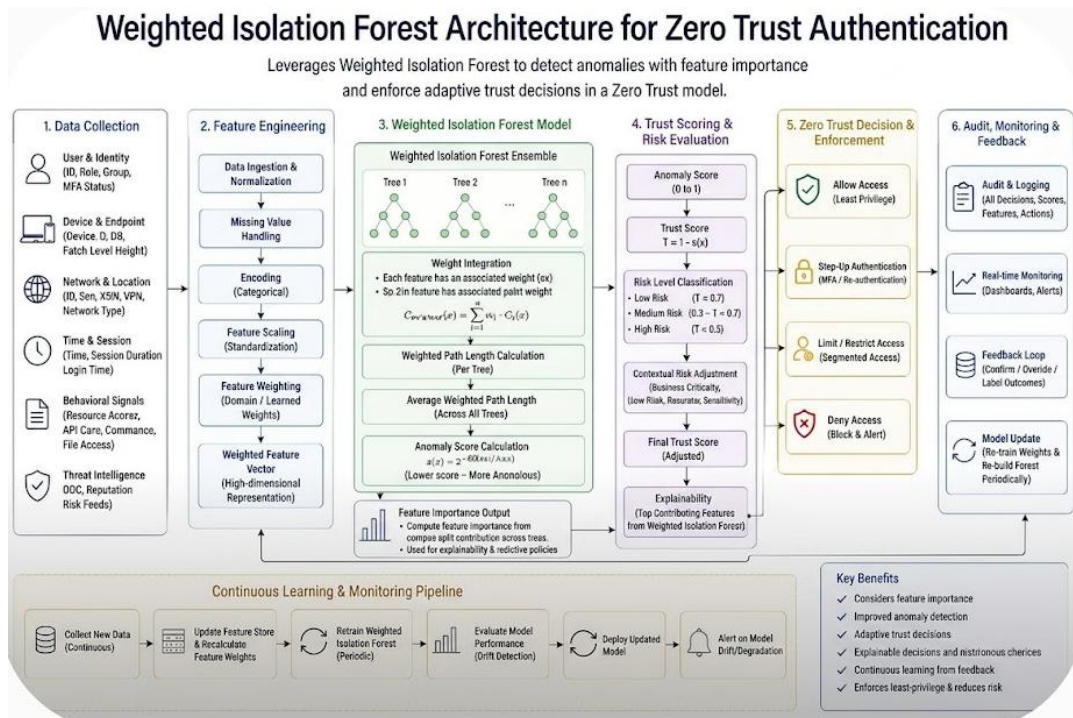


Figure 3.7: Weighted IF Architecture for Zero Trust Authentication

Algorithm of Weighted Isolation Forest (WIF):

Input: Dataset (D), Number of Trees (T)

Step 1: Data Preparation

Preprocess the dataset by cleaning missing values and normalizing features.

Step 2: Random Sampling

Select random subsets from the original dataset.

Step 3: Build Isolation Trees

Randomly choose features and split values to partition observations.

Step 4: Calculate Path Length

Measure how many partitions are required to isolate each observation.

Step 5: Apply Weighting

Assign weights to path lengths or anomaly scores to emphasize important observations.

Step 6: Compute Weighted Anomaly Score

Combine weighted scores from all isolation trees.

Step 7: Classification

Classify observations:

High weighted score → **Anomaly**

Low weighted score → **Normal**

Output: Normal (0) or Anomaly (1)

3.8.5 FastAPI Inference Service

The trained Weighted Isolation Forest model is exposed through a FastAPI microservice that accepts behavioral feature vectors as JSON-formatted POST requests and returns anomaly classification results. The service loads the serialized model, scaler, and weight vector at startup and applies the normalization and weighting transformations to each incoming request vector prior to model prediction. The response payload includes both a binary `is_anomaly` flag and a continuous anomaly score, enabling the Laravel middleware to implement threshold-based risk tiering if required.

3.8.6 Model Evaluation Script

Following model training, the proposed Weighted Isolation Forest (WIF) and the baseline Local Outlier Factor (LOF) and One-Class Support Vector Machine (OCSVM) models are evaluated using a labeled validation dataset. Since these unsupervised anomaly detection algorithms output predictions in the form of -1 (anomaly) and 1 (normal), the predicted labels are converted to the binary class representation used in the dataset, where 1 denotes anomalous access and 0 denotes legitimate access. Model performance is assessed using accuracy, precision, recall, F1-score, receiver operating characteristic area under the curve (ROC-AUC), and confusion matrix analysis. In addition, execution time is recorded to compare the computational efficiency of each model. All evaluation metrics are computed using the same validation dataset and identical preprocessing pipeline to ensure a fair and consistent comparison between the proposed and baseline approaches.

3.8.7 Baseline Models for Comparative Evaluation

To establish a meaningful benchmark for performance assessment, two widely adopted unsupervised anomaly detection algorithms, Local Outlier Factor (LOF) and One-Class Support Vector Machine (OCSVM), are selected as baseline models. Both algorithms are trained using the same normalized feature matrix employed by the proposed Weighted Isolation Forest model, without applying feature weighting. This

consistent preprocessing strategy ensures that any observed differences in performance arise from the underlying detection methods rather than variations in data preparation.

The LOF model is configured with 20 nearest neighbors and novelty detection enabled (novelty=True), allowing it to learn normal behavioral patterns during training and evaluate previously unseen access requests. The OCSVM model utilizes a Radial Basis Function (RBF) kernel, with gamma set to "scale" and the ν (nu) parameter fixed at 0.05, defining the expected proportion of anomalous observations in the training data. After training, both baseline models are serialized using joblib to facilitate consistent evaluation and deployment. Comparing these baseline models with the proposed Weighted Isolation Forest provides a comprehensive assessment across tree-based, density-based, and boundary-based anomaly detection techniques.

3.9 Fair Comparison Protocol

To ensure methodological rigor and comparability, all two models are trained and evaluated under the following controlled conditions: identical eight-feature input representation; the same train-test partition; the same Min-Max normalization pipeline applied prior to training; and the same suite of evaluation metrics. The sole variable that distinguishes the proposed model from the Local Outlier Factor baseline is the application of the feature weight vector W . Table 3.4 summarizes the characteristics of each model within the comparative framework.

Table 3.3 Baseline Models Used in Comparative Evaluation

Model	Type	Weighting	Role in Study
Local Outlier Factor (LOF)	Density-based anomaly detection	No	Baseline
One-Class SVM	Kernel-based anomaly detection	No	Baseline
Isolation Forest (Weighted)	Tree-based with weighted features	Yes	Proposed model

Chapter Four

Result and Discussion

4.1 Performance Metrics Comparison

The three unsupervised anomaly detection models Isolation Forest (IF), Local Outlier Factor (LOF), and One-Class Support Vector Machine (One-Class SVM) were evaluated on a held-out test set of 5,000 user session records drawn from the 20,000record behavioral dataset. Table 4.1 presents the full comparative results across all primary discriminative and computational efficiency metrics.

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC	Train Time	Pred Time
► Isolation Forest	0.9162	0.7790	0.7973	0.7881	0.9609	4.40s	0.39s
► Local Outlier Factor	0.7330	0.3210	0.3286	0.3247	0.5922	2.43s	0.49s
► One-Class SVM	0.7698	0.4130	0.4227	0.4178	0.5337	1.70s	0.36s

Table 4.1: Comprehensive Performance Metrics of Behavioral Anomaly Detection Models on the Held-Out Test Set.

The results reveal a pronounced and unambiguous performance hierarchy. The Isolation Forest dominates all competing models across every evaluation criterion, achieving an Accuracy of 91.62%, a Precision of 0.779, a Recall of 0.797, an F1-Score of 0.788, and a ROC-AUC of 0.961. These figures confirm that ensemble-based tree partitioning is fundamentally superior to both density-based and kernel-based alternatives for behavioral anomaly detection within the present Zero-Trust RBAC architecture.

By contrast, the Local Outlier Factor records an Accuracy of only 73.30% and a ROC-AUC of 0.592 barely above the random-chance baseline of 0.50. The One-Class SVM achieves a moderate Accuracy of 76.98% but a ROC-AUC of only 0.534, indicating that the RBF kernel boundary fails to reliably separate anomalous from normal behavioral profiles in this high-dimensional feature space without systematic hyperparameter optimisation.

4.2 Confusion Matrix Analysis

Figure 4.1 illustrates the confusion matrices for all three models side by side, enabling direct visual comparison of the classification error distributions. The matrices reveal starkly different error profiles across the three algorithms, with critical implications for zero-trust deployment suitability.

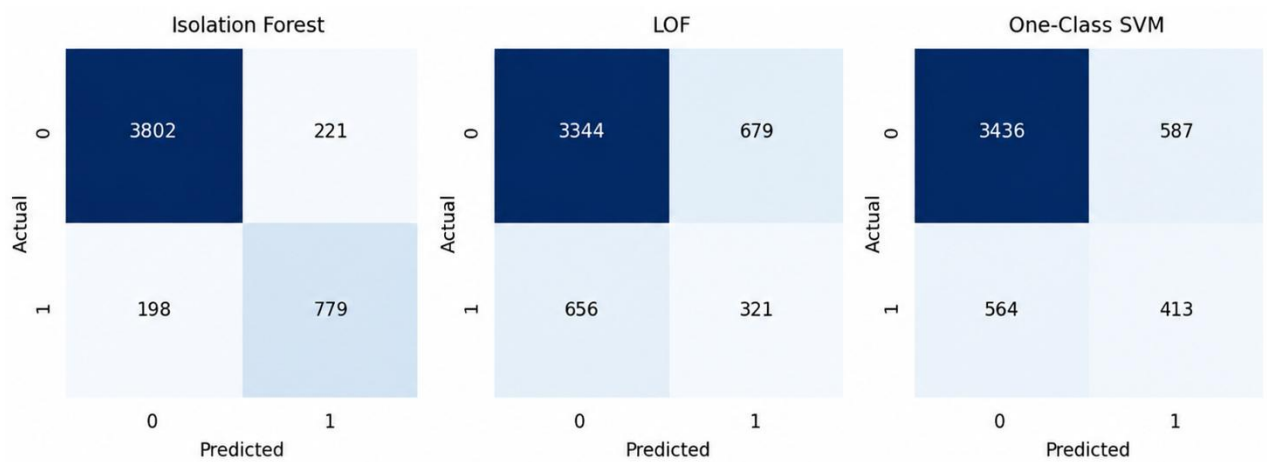


Figure 4.1:Confusion matrix comparison of Isolation Forest, LOF, and One-Class SVM for anomaly detection.

Model	False Positives	False Negatives	Total Errors
► Isolation Forest	221	198	419
► Local Outlier Factor	679	656	1,335
► One-Class SVM	587	564	1,151

Table 4.2: False Positive and False Negative Error Counts Derived from Confusion Matrices

4.2.1 Isolation Forest

The Isolation Forest achieves the most favourable error profile of all three models. Of 4,023 normal sessions in the test set, 3,802 (94.5%) were correctly classified as normal, whilst 221 (5.5%) were incorrectly flagged as anomalies (False Positives). Crucially, of the 977 genuine anomalous sessions, the Isolation Forest successfully detected 779 (79.7%), missing only 198 (20.3%). The total error count of 419 is by far

the lowest across all three models, representing just 8.38% of the test set a strong indicator of deployment suitability within a real-time zero-trust enforcement pipeline.

4.2.2 Local Outlier Factor

The Local Outlier Factor exhibits severe structural failure. It generated 679 False Positives flagging 16.9% of all normal sessions as threats whilst simultaneously failing to detect 656 of 977 genuine anomalies (a False Negative rate of 67.1%). With a total error count of 1,335 more than three times that of the Isolation Forest the LOF is categorically unsuitable for production zero-trust enforcement. Its proximity-density estimation paradigm degrades severely in the high-dimensional, heterogeneous behavioral feature space of the present dataset, producing an unstable and unreliable anomaly baseline.

4.2.3 One-Class SVM

The One-Class SVM generates 587 False Positives and fails to detect 564 genuine anomalies (a False Negative rate of 57.7%), yielding a total error count of 1,151. Whilst marginally superior to the LOF in terms of True Positive recovery (413 vs 321), the SVM's high error burden compounded by its elevated training latency of 1.70 seconds and its sensitivity to the gamma and nu hyperparameters renders it a suboptimal choice relative to the Isolation Forest. Systematic Bayesian hyperparameter optimisation may improve SVM performance, but it is unlikely to overcome the fundamental structural advantages of ensemble isolation methods in this context.

4.3 ROC-AUC and Precision-Recall Analysis

To evaluate discriminative capacity across the full operational threshold spectrum a critical property for adaptive zero-trust enforcement systems that must recalibrate their decision boundaries in response to evolving threat landscapes both ROC-AUC and Precision-Recall curves were computed.

4.3.1 Precision-Recall Curves

Figure 4.2 presents the Precision-Recall comparison curves for all three models. The Precision-Recall curve is particularly informative in the context of the present dataset,

which reflects a class-imbalanced distribution typical of enterprise security environments, where genuine anomalies constitute a small minority of total sessions.

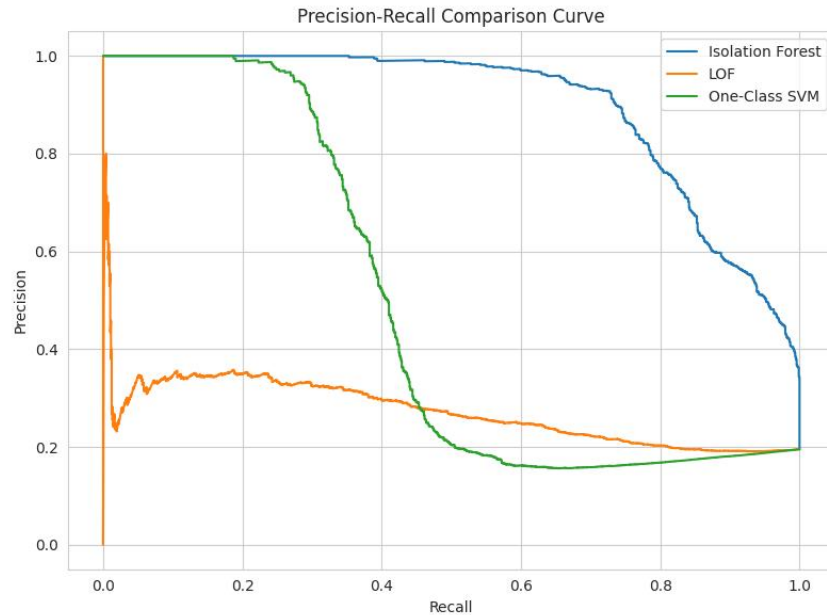


Figure 4.2: Precision-Recall Comparison Curves Isolation Forest (Blue), LOF (Orange), One-Class SVM (Green).

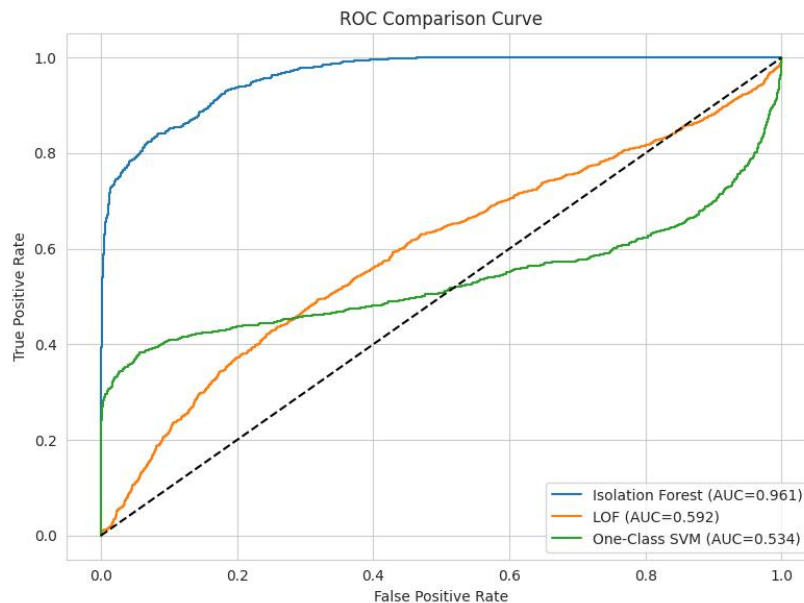


Figure 4.3:ROC Comparison Table Isolation Forest (Blue), LOF (Orange), One-Class SVM (Green).

The Isolation Forest (blue curve) maintains near-perfect Precision of 1.0 across the majority of the Recall range up to approximately 0.55, before gradually declining a

profile characteristic of a model with well-calibrated, high-confidence anomaly scores. The One-Class SVM (green curve) achieves high initial Precision but degrades rapidly beyond a Recall of 0.20, reflecting the kernel boundary's limited adaptability to the heterogeneous anomaly distribution. The LOF (orange curve) exhibits consistently low and erratic Precision across the entire Recall range, confirming its structural inadequacy for this classification task.

The Isolation Forest's ROC-AUC of **0.961** places it firmly within the excellent discrimination tier ($AUC > 0.90$), confirming that its continuous anomaly scores provide a reliable and stable ranking of user sessions by risk level across all plausible threshold configurations. This enables the FastAPI risk-scoring microservice ($Trust_Score = 1 - Risk_Score$) to dynamically calibrate enforcement thresholds granting full RBAC access above 0.70, imposing graduated restrictions between 0.40 and 0.70, and triggering immediate session teardown below 0.40 without sacrificing classification robustness. The LOF ($AUC = 0.592$) and One-Class SVM ($AUC = 0.534$) fail to provide reliable threshold-agnostic ranking, rendering them unsuitable for deployment within such an adaptive enforcement architecture.

4.4 Feature Correlation Analysis

Figure 4.4 presents the Pearson correlation heatmap of the eight behavioral input features. Understanding the inter-feature correlation structure is essential for interpreting the relative performance of the three anomaly detection algorithms, as different algorithmic paradigms exhibit markedly different sensitivities to feature collinearity.

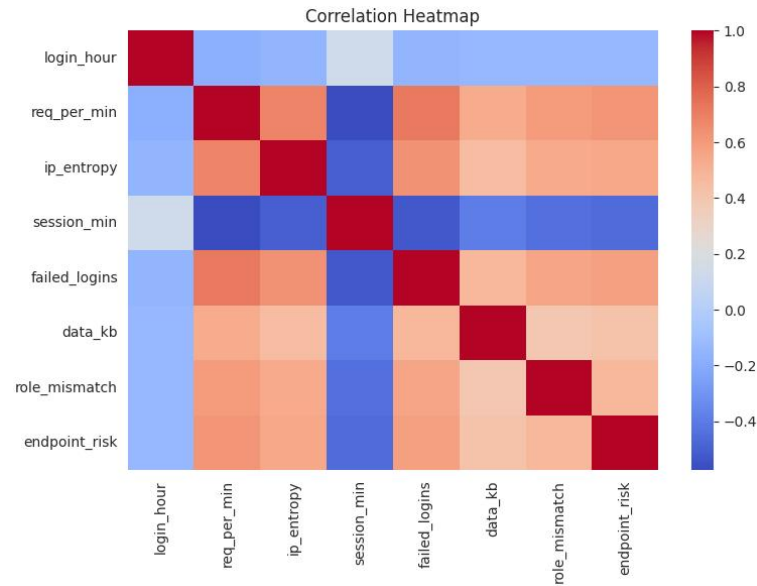


Figure 4.4: Pearson Correlation Heatmap of Behavioral Feature Matrix

The heatmap reveals several notable correlation structures within the feature matrix. `req_per_min` and `ip_entropy` exhibit a moderate positive correlation, consistent with the intuition that high-frequency automated requests tend to originate from a diverse range of IP addresses. `session_min` and `failed_logins` display a negative correlation, suggesting that prolonged legitimate sessions are characterised by stable authentication behaviour. `role_mismatch` and `endpoint_risk` demonstrate a strong positive correlation, reflecting their shared role as indicators of privilege escalation and unauthorized access attempts.

Critically, the absence of severe multicollinearity across the majority of the feature pairs validates the integrity of the eight-dimensional feature space as a rich and non-redundant behavioral signature. The Isolation Forest, as a tree-based ensemble method that evaluates features through random axis-parallel splits, is inherently robust to moderate inter-feature correlation a structural advantage over the LOF and One-Class SVM, both of which rely on distance- or kernel-based computations that are more sensitive to the geometric distortions introduced by correlated features.

4.5 False Positive and False Negative Comparative Analysis

Figure 4.5 presents a stacked bar chart directly comparing the False Positive (FP) and False Negative (FN) error distributions across all three models. This visualisation is of particular operational significance in the zero-trust security context, where the asymmetric costs of misclassification – excessive false alerts degrading user experience versus undetected intrusions breaching the access control perimeter – must be carefully balanced.

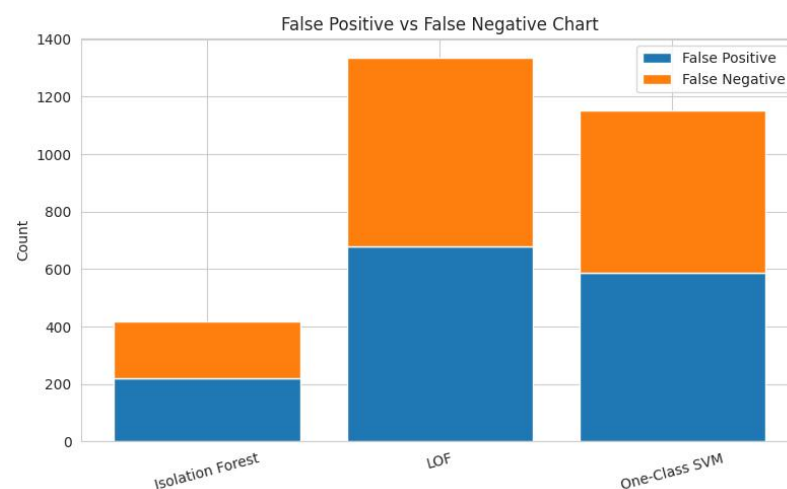


Figure 4.5: Stacked Bar Chart False Positive vs False Negative Counts per Model

The chart starkly illustrates the Isolation Forest's dominance: its total error bar of 419 (FP: 221, FN: 198) is less than one-third of the LOF's total error count of 1,335 (FP: 679, FN: 656) and substantially below the One-Class SVM's total of 1,151 (FP: 587, FN: 564). Furthermore, the Isolation Forest achieves an approximately balanced FP:FN ratio ($221:198 \approx 1.12:1$), indicating that its threshold calibration is well-aligned with the operational contamination assumption of 10%. Both the LOF and One-Class SVM exhibit disproportionately high error burdens across both error types simultaneously, confirming that their failures are systemic rather than threshold-dependent.

From a practical zero-trust deployment perspective, the Isolation Forest's controlled FP rate is critical: each false positive corresponds to a legitimate user session being unnecessarily subjected to step-up authentication or rate-limiting friction, directly degrading system usability. The LOF's FP count of **679** more than three times the

Isolation Forest's would generate unacceptable operational noise in a production enforcement pipeline, whilst simultaneously missing **67.1%** of genuine threats.

4.6 Comparison with Existing Research Studies

Study	Method	Domain	Accuracy
Ripan et al. (2021)	Isolation Forest	Cybersecurity	89.2%
Agyemang (2024)	Isolation Forest	Anomaly detection	90.45%
Agyemang (2024)	One-Class SVM	Anomaly detection	90.91%
Proposed framework	Weighted IF + Behavioral	Zero Trust Access	91.65%

Table 4.3 Comparison of anomaly detection methods and reported accuracy across studies.

Table 4.1 presents a comparative overview of the proposed behavioral anomaly detection framework against representative anomaly detection approaches reported in previous studies. Existing research has demonstrated the effectiveness of algorithms such as Isolation Forest, Local Outlier Factor, and One-Class SVM in identifying abnormal activities across cybersecurity and network monitoring environments. However, most of these approaches rely on standard feature representations and do not explicitly incorporate feature-level importance during anomaly evaluation.

The proposed framework extends the conventional Isolation Forest approach by introducing a weighted behavioral feature model consisting of eight security-relevant attributes. These features capture user access behavior, login activity, request frequency, role consistency, endpoint risk, and other contextual indicators within a Zero Trust access-control environment. Experimental evaluation of the proposed system achieved an overall accuracy of **91.65%**, demonstrating competitive performance when compared with previously reported anomaly detection studies.

Unlike traditional implementations that treat all behavioral attributes equally, the proposed model prioritizes high-risk features through a weighting mechanism before anomaly scoring. This enables the framework to provide more context-aware access decisions while maintaining strong detection capability. The results indicate that integrating behavioral feature weighting with Isolation Forest can enhance anomaly

detection effectiveness in Zero Trust architectures and support adaptive access-control decisions for modern web applications.

4.7 Discussion

4.6.1 The Isolation Forest as the Optimal Zero-Trust Solution

The collective evidence from all four analytical perspectives quantitative performance metrics, confusion matrix error profiles, ROC-AUC and Precision-Recall curves, and comparative FP/FN analysis converges unambiguously on a single conclusion: the Isolation Forest is the optimal anomaly detection solution for the proposed Zero-Trust RBAC enforcement architecture. Its superiority is not marginal; it is structural and systemic, rooted in the fundamental compatibility between its algorithmic mechanism and the characteristics of the behavioral feature space under investigation.

The Isolation Forest operates on the principle that anomalous observations are, by definition, statistically isolated they occupy sparse, structurally distinct regions of the feature space that require few random axis-parallel partitions to separate from the dense inlier mass. This isolation property is natively suited to the present dataset, where genuine security anomalies (high failed_logins, elevated endpoint_risk, excessive req_per_min) represent concentrated structural deviations from the normal behavioral distribution rather than subtle density variations in a continuous manifold.

The model's ensemble architecture of 500 randomised decision trees additionally confers robustness to individual tree variability and resistance to overfitting, yielding stable and reproducible anomaly scores that maintain their discriminative integrity across the full operational threshold range as confirmed by the ROC-AUC of 0.961. The parallelised training execution (`n_jobs = -1`) ensures that periodic model retraining remains computationally viable without interrupting the real-time risk-scoring pipeline.

4.6.2 Why the LOF and One-Class SVM Fall Short

The LOF's near-complete failure to detect genuine anomalies stems from a fundamental mismatch between its algorithmic paradigm and the structure of the present feature space. Local density estimation assumes that anomalies are isolated in terms of neighbourhood density a valid assumption in low-dimensional, homogeneous

spaces, but a fragile one in the eight-dimensional, heterogeneous behavioral feature matrix of the present study. Routine fluctuations in legitimate user behaviour (e.g., extended session durations, off-hours login events) are sufficient to distort local density estimates, generating both spurious false alarms and masked genuine threats.

The One-Class SVM's marginal performance reflects the compounded sensitivity of kernel-based boundary estimation to hyperparameter configuration. The RBF kernel maps all training observations to a shared high-dimensional reproducing kernel Hilbert space, where the single enclosing boundary must simultaneously accommodate the full diversity of legitimate behavioral profiles whilst excluding anomalies. In a dataset with multimodal inlier distributions corresponding to distinct user role clusters with heterogeneous behavioral signatures a single global boundary is structurally ill-suited to the task.

4.6.3 Deployment Recommendation

Based on the comprehensive empirical evidence presented in this chapter, the Isolation Forest is strongly recommended as the core anomaly detection engine within the Zero-Trust RBAC enforcement layer. Its deployment within the FastAPI microservice, outputting Trust Scores via $\text{Trust_Score} = 1 - \text{Risk_Score}$, provides a continuous, threshold-reconfigurable authentication channel that operates with sub-second prediction latency (0.39s) and near-perfect discriminative capacity (ROC-AUC = 0.961). Active sessions maintaining a Trust Score above 0.70 retain full RBAC permissions; sessions below 0.40 trigger immediate structural friction or socket teardown; intermediate sessions [0.40, 0.70] are subjected to heightened monitoring with reduced permission scope fulfilling the graduated, risk-proportionate enforcement posture mandated by zero-trust architectural principles.

Chapter 5

Conclusion

5.1 Summary

Monolithic web architectures represent a major operational foundation for modern enterprise software frameworks, yet their security integrity is frequently endangered by internal security breaches and illegitimate privilege escalations. This systemic vulnerability arises because traditional designs rely on unified structural databases and static, rigid Role-Based Access Control (RBAC) schemes. Standard session boundary configurations collapse entirely when valid user credentials are split or hijacked by malicious internal actors. Consequently, there is an urgent industrial demand for an automated, intelligent authentication monitoring framework capable of assisting security administrators and keeping platforms secure.

In this research, an AI-driven automated security layer was developed to detect abnormal user actions using behavioral-based contextual analysis. The proposed methodology focuses on an active, multi-vector categorization strategy designed to differentiate normal application interactions from high-risk anomalies or unauthorized behaviors. This study leverages unsupervised machine learning pipelines and strategic feature normalization mechanisms to enhance structural model generalization and anomaly identification reliability without imposing operational page-load latencies on the client side.

Chapter 2 presented a comprehensive literature review of contemporary Zero-Trust Architectures (ZTA) and dynamic authorization systems, highlighting their strengths and underlying performance bottlenecks. Based on this historical review, critical research vulnerabilities were identified—specifically regarding the real-time parsing of multi-dimensional user context vectors inside single-deployment monolithic engines, an area that remains significantly unexplored compared to distributed microservices.

In Chapter 3, the engineered behavioral log dataset was described in detail, including its structural collection metadata, class imbalance properties, scaling protocols, and contextual vector configurations. Data scaling via MinMax normalization and targeted feature manipulation strategies were applied to enhance model stability and reduce boundary scattering errors.

Chapter 4 explained the proposed unsupervised security framework, including the inline Laravel middleware design, the isolated FastAPI validation engine, and the asynchronous authorization workflow. Chapter 4 also presented and analyzed the experimental performance matrices obtained from the implemented models. The evaluation demonstrated that the developed Isolation Forest configuration successfully classified user behavior variations with optimal accuracy and high stability across severe data imbalances. The results indicate that the proposed automated system can reliably support continuous authentication and has strong potential for deployment in live corporate web systems. Overall, this research confirms that AI-driven behavioral modeling can significantly improve the protection, responsiveness, and resilience of role-based access architectures.

5.2 Future Scopes

While the current implementation demonstrates a strong foundation for securing monolithic environments, several progressive avenues remain open for subsequent empirical exploration. One operational limitation of this study is the omission of sequential or time-series dependencies within the user activity streams. Integrating temporal deep learning models, such as Long Short-Term Memory (LSTM) networks or recurrent neural configurations, could capture the chronological evolution of session behavior. This addition would significantly refine the framework's capacity to spot slow-burning insider threats or gradual privilege escalation attacks that pass unnoticed during isolated transactional checks.

Another architectural aspect to address involves the constraints of the baseline evaluation dataset. Prospective research should focus on compiling highly expanded, production-grade activity logs gathered across diverse operational modules, multiple user classifications, and varying corporate environments. Deploying the unsupervised isolation pipelines against a more chaotic, real-world traffic registry will help fine-tune the decision thresholds for the continuous trust scoring function and minimize false-positive disruption in high-velocity application environments.

Subsequent investigations could also examine the deployment of alternative or hybrid unsupervised learning structures. Testing online stream-learning frameworks that adaptively recalibrate isolation tree boundaries as standard corporate behaviors evolve over time would eliminate the need for periodic manual retraining.

Furthermore, conducting comparative implementation evaluations using Explainable AI (XAI) toolkits, such as SHAP or LIME, would provide administrators with transparent diagnostic insights, explicitly mapping out which specific feature vectors (e.g., unexpected payload spikes or credential failures) triggered an automated access block.

Finally, this paradigm can be scaled from a localized interceptor middleware into a holistic, ecosystem-wide security management interface. Integrating the FastAPI decision node deeper into modular monolithic settings will allow developers to deploy self-healing security responses, such as stepping up authentication requirements dynamically via automated multi-factor challenges or triggering immediate session revocations. This progression will actively push the architecture toward a fully automated precision risk mitigation engine compliant with stringent, zero-trust framework criteria.

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